

Lab 2: Digital Differential Analyzer (BLA)

Adding graphics package in Dev C++

1. Download the Dev C++ also download <https://tinyurl.com/3ve35mzh>
2. You need to copy “graphics.h” and “winbgim.h” into include the directory of Dev-Cpp program resource files. The exact directory address is –
C:\Program Files (x86)\Dev-Cpp\MinGW64\include
3. Next, copy “libbgi.a” to the lib folder, which should be inside MinGW64. The directory address is –
C:\Program Files (x86)\Dev-Cpp\MinGW64\lib
4. In Dev C++, Set it to the latest **TDM - GCC [version] 32-bit release.**
5. In Dev C++, Expand tools and select Compiler Options.
 - Ensure the “Add the following commands when calling the linker” check box is selected.
 - Then add these linkers in the input box – (copy and paste this line)
-static-libgcc -lbgi -lgdi32 -lcomdlg32 -luuid -oleaut32 -ole32
 - Then click on OK to save.

LAB SHEET

Theory: Briefly explain BLA algorithm with its algorithm

Lab works:

Question 1: Write a code to draw a line using the BLA algorithm.

```
#include<stdio.h>

#include<graphics.h>

int drawline(int x0, int y0, int x1, int y1)
{
    int dx, dy, p, x, y;
    dx=x1-x0;
    dy=y1-y0;
    x=x0;
    y=y0;
```

```

    p=2*dy-dx;
while(x<x1)
{
    if(p>=0)
    {
        putpixel(x,y,7);
        y=y+1;
        p=p+2*dy-2*dx;
    }
    else
    {
        putpixel(x,y,7);
        p=p+2*dy;
    }
    x=x+1;
}
}

int main()
{
    int gdriver=DETECT, gmode, error, x0, y0, x1, y1;
    initgraph(&gdriver, &gmode, NULL);
    printf("Enter co-ordinates of first point: ");
    scanf("%d%d", &x0, &y0);
    printf("Enter co-ordinates of second point: ");
    scanf("%d%d", &x1, &y1);

```

```
drawline(x0, y0, x1, y1);  
getch();  
return 0;  
}
```

Conclusion

- What did u learn?